

# SMOKE

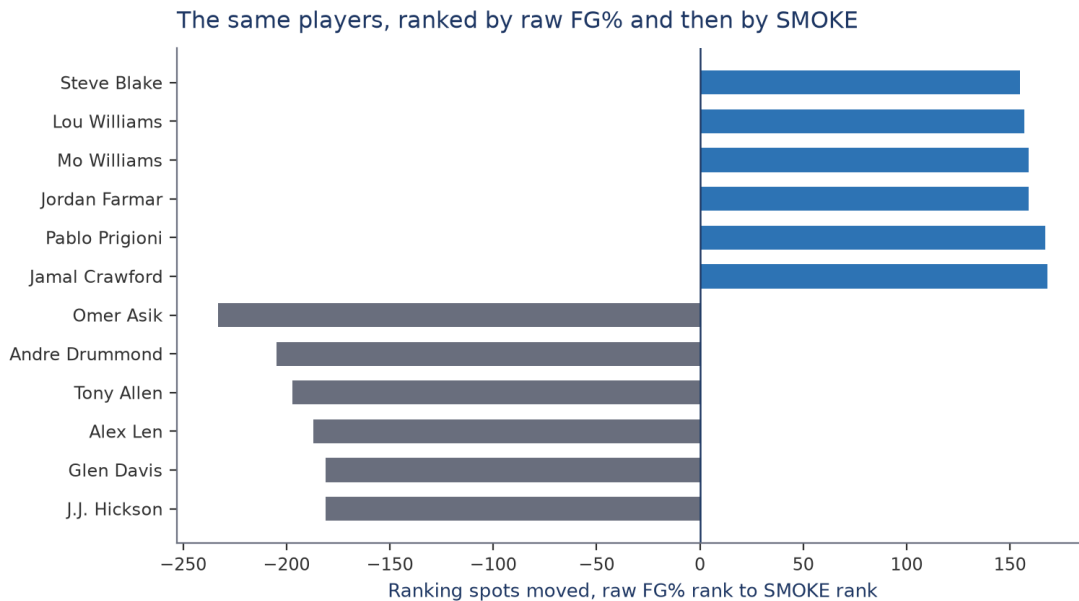
*Shots Made Over Known Expectation. Shot-making, separated from shot selection.*

## The problem

Field goal percentage rewards two different things at once: taking easy shots and making hard ones. Two players can post the same percentage while one lives on open dunks and the other makes contested jumpers, and the box score cannot tell them apart.

## What SMOKE measures

A model estimates every shot's make probability from its difficulty: distance, defender distance, shot clock, and touch time, across 128,069 tracked shots. SMOKE is a player's actual makes minus the makes the model expected. It isolates shot-making skill from shot selection. Stephen Curry led 2014-15 at 62 makes above expectation, with a confidence range of 32 to 91; his number is skill, not luck.



*Risers make difficult shots that raw percentages undervalue. Fallers post high percentages on shots the model expected them to make anyway. Both lists are explainable, which is the point.*

## Why trust this number

- It repeats. Year-over-year stability is 0.48, higher than effective field goal percentage at 0.43, and it predicts next season's shot-making better than efficiency does (0.40 versus 0.31).
- It is honest about noise. Every player's value ships with a confidence interval and a reliability weight, and small samples are shrunk toward average. No competing shot-quality product publishes either.
- It is fair and robust. No playing style is structurally penalized (interior big men included). Opponent quality and venue explain just 0.6% of the variance in SMOKE, and rankings before and after controlling for them correlate at a Spearman correlation of 0.996.